Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 3^{rd} Semester End-Semester Examination, November 2023
Discrete Structures (CS - 2101)

Time: 3 hours **Full Marks: 50**

Answer Q1 and *any* three from the rest.
Extra attempts will NOT be evaluated.
Write all parts of the same question together.
Marks will be deducted if answers are vague and intermediate steps are skipped

1. (a) Validity of "proof by induction" can be proven using the __________.
   (b) Explain which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$: (i) $\{2, 4, 6\}, \{1, 3, 5\}$, (ii) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$, (iii) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$.
   (c) Let $Graph(x)$ be predicate which denotes "x is a graph." Let $Connected(x)$ be a predicate, which denotes "x is connected". Write the logical representation of the proposition: "Not every graph is connected."
   (d) A proof that $p \to q$ is true based on the fact that q is true, is known as __________.
   (e) State *True* or *False*?
      i. An equivalence relation has reflexive, anti-symmetric, and transitive properties.
      ii. A partially ordered relation has reflexive, symmetric, and transitive properties.
   (f) Two sets A and B contain a and b elements, respectively. If the power set of A contains 16 more elements than that of B, what are the values of a and b?
   (g) How many reflexive, symmetric, and anti-symmetric relations are possible in a set containing n elements?
   (h) Show that in any set of six classes, each meeting regularly once a week on a particular day of the week, there must be two that meet on the same day, assuming that no classes are held on weekends.
   (i) Let a poset is represented by the Hasse diagram in Figure 1. Find all upper bounds of $\{a, b, c\}$ and lower bounds of $\{f, g, h\}$.

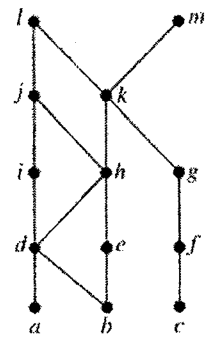

Figure 1: Hasse Diagram

$[1 + 3 + 2 + 1 + 2 + 2 + 3 + 2 + (2 + 2) = 20]$

2. (a) Define transitive closure of a relation R in terms of connectivity relation.
   (b) Write a pseudo code for Warshall's Algorithm to find the transitive closure of a set of n elements. Find out the running time complexity for this algorithm in terms of the input size n.
   (c) Use the above algorithm to find the transitive closure of the relation $\{(2, 1), (2, 3), (3, 1), (3, 4), (4, 1), (4, 3)\}$ on the set $\{1, 2, 3, 4\}$.

$[1 + (3 + 2) + 4 = 10]$

3. (a) What is a lattice? Determine whether the poset $S = (\{1, 3, 6, 9, 12\}, |)$ is a lattice.

   (b) In a company, the lattice model of information flow is used to control sensitive information with security classes represented by ordered pairs (A, C). Here, A is an authority level, which may be *non-proprietary (0)*, *proprietary (1)*, *restricted (2)*, or *registered (3)*. A category C is a subset of all projects $\{Cheetah, Impala, Puma\}$. (Names of animals are often used as code names for projects in companies.)
   Answer the following [give justifications]:
   (i) Is the information flow $(Restricted, \{Cheetah\}) \rightarrow (Registered, \{Cheetah, Impala\})$ permitted?
   (ii) Into which classes is information from $(Proprietary, \{Cheetah, Puma\})$ permitted to flow?

   (c) Find a topological sorting for the poset $A = (\{1, 2, 4, 5, 10, 15, 20\}, |)$ that differs from the ordinary "less than or equal to" relation $\leq$. $[(1+2)+(2+2)+3=10]$

4. (a) Use proof by contraposition to show if n is an integer and n^3+5 is odd, then n is even.

   (b) Prove by contradiction that the square root of an irrational number is irrational.

   (c) Use mathematical induction to prove the following generalization of one of De Morgan's laws:

$$\overline{\bigcup_{j=1}^{n} A_j} = \bigcap_{j=1}^{n} \overline{A_j}$$

   whenever $A_1, A_2, \ldots, A_n$ are subsets of an universal set U and $n \geq 2$. $[3+3+4=10]$

5. (a) Deduce the expression for the *Generalized Pigeonhole Principle* considering there are N objects and k boxes, and when these objects are distributed among the boxes, the minimum number of objects to be placed in one of the boxes is at least r.

   (b) Let R be the relation on the set of all people who have visited a particular Web page such that xRy if and only if person x and person y have followed the same set of links starting at this Web page (going from Web page to Web page until they stop using the Web). Show that R is an equivalence relation.

   (c) Determine whether the relation represented in Figure 2 has reflexive, symmetric, antisymmetric, and transitive properties (give brief justification). $[3+3+4=10]$

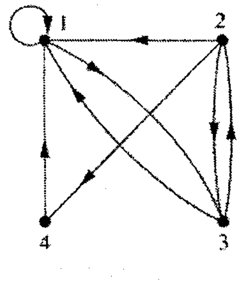

Figure 2: Directed Relation Graph